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$$I = \int_0^1 \frac{xe^x}{(x+1)^2} dx$$

Partiële integratie:

$$\begin{aligned} \begin{cases} u = xe^x \\ dv = \frac{1}{(x+1)^2} dx \end{cases} &\Rightarrow \begin{cases} du = (e^x + xe^x) dx \\ v = -\frac{1}{x+1} \end{cases} \\ I &= \left[ -\frac{xe^x}{x+1} \right]_0^1 - \int_0^1 \left( -\frac{1}{x+1} \right) e^x(1+x) dx \\ &= \left[ -\frac{xe^x}{x+1} \right]_0^1 + \int_0^1 e^x dx \\ &= \left[ -\frac{xe^x}{x+1} + e^x \right]_0^1 \\ &= \left[ \frac{e^x}{x+1} \right]_0^1 \\ &= \frac{e}{2} - 1 \end{aligned}$$

## References